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P525/3
CHEMISTRY
(PRACTICAL)
PAPER 3
NOV. 2020
3 ¼ HOURS

Uganda Advanced Certificate of Education
CHEMISTRY PRACTICAL
PAPER 3
3 HOURS 15 MINUTES

Answer all questions in the spaces provided.

Candidates are not allowed to start working with the apparatus for the first 15 minutes. This time is to enable candidates to read the question paper and make sure they have all the apparatus and chemicals that they may need.

FOR EXAMINER'S USE ONLY			
Q.1	Q.2	Q.3	TOTAL

1. You are provided with the following

FA₁ which is potassium manganate(VII) of unknown concentration

FA₂ which is hydrogen peroxide

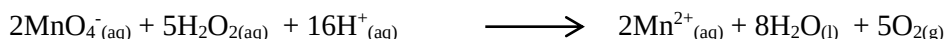
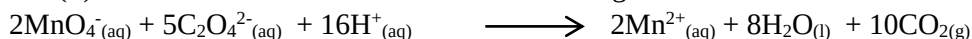
Solid **Q** which is sodium ethanedioate

2M sulphuric acid

You are required to

(i) Standardize **FA₁** using **Q**

(ii) Use **FA₁** to determine the volume strength of **FA₂**



PROCEDURE

PART I:

- (a) Weigh accurately 1.3g of **Q** into a beaker and then add about 100cm³ of distilled water. Transfer the solution into a 250cm³ volumetric flask and make up to the mark with distilled water. Label this solution **FA₃**
- (b) Pipette 25.0cm³ (or 20.0cm³) of **FA₃** into a conical flask followed by an equal volume of 2M sulphuric acid. Heat the mixture to about 60°C. Titrate the hot solution with **FA₁** from the burette until the solution just turns pink. Repeat the titration until you obtain consistent results. Record your results in the table below.

Mass of **Q** + beakerg

Mass of empty beakerg

Mass of **Q** usedg

Volume of pipette usedcm³

Experiments	1	2	3
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of FA₁ used (cm ³)			

Record values used to calculate the average volume of **FA₁** used

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Average volume of **FA₁** used

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Questions:

- (c) Calculate the concentration of manganate(VII) ions in **FA₁** per litre of solution
(Na=23, C=12, O=16)

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PART II

Using a measuring cylinder transfer 10cm³ of **FA₂** into a 250cm³ volumetric flask. Add distilled water up to the mark and label this solution **FA₄**.

Pipette 25.0cm³ (or 20.0cm³) of **FA₄** into a conical flask followed by an equal volume of 2M sulphuric acid. Titrate the mixture with **FA₁** from the burette until the solution just turns pink. Repeat the titration until you obtain consistent results and record your results in the table below.

Volume of pipette usedcm³

Experiments	1	2	3
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of FA₁ used (cm ³)			

Record the titre values used to calculate average volume of **FA₁** used

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Average volume of **FA₁** used

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Questions:

(a) Calculate the

(i) number of moles of MnO_4^- that reacted with H_2O_2 in **FA₁**

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(ii) moles of H_2O_2 in 250cm^3 of **FA₄**

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(iii) molarity of **FA₂**

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(iv) percentage of hydrogen peroxide in **FA₂**

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(v) volume strength of **FA₂**

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2. You are provided with substance **X** which contains two cations and two anions. You are required to carry out the following tests on **X** and identify the anions and cations in **X**. Record your observations in the table below.

TESTS	OBSERVATIONS	DEDUCTIONS
(a) Heat a spatula endful of X in a dry test tube.	Green powdery solid Black residue Yellow residue when hot turns white on cooling Colourless gas turn moist blue litmus paper red and lime water milky	Fe ²⁺ Cu ²⁺ Ni ²⁺ Cr ³⁺ FeO ,CuO, NiO, CrO, hence Fe ²⁺ Cu ²⁺ Ni ²⁺ Cr ³⁺ ZnO hence Zn ²⁺ CO ₂ from CO ₃ ²⁻ , C ₂ O ₄ ²⁻ CH ₃ COO-

	White fumes with a sweet smell forms a yellow precipitate with Brady's reagent Colourless liquid turns anhydrous copper (II) sulphate from white to blue White sublimate	CH_3COCH_3 vapor from CH_3COO^- Hydrated salt CH_3COO^-
(b) Place 2 spatula endfuls of X in a test tube, add 3 drops of concentrated sulphuric acid and heat.	Dense white fumes with sharp vinegar smell turn blue litmus paper red Colourless gas turn moist blue litmus paper red and lime water milky	CH_3COOH vapour from CH_3COO^- CO_2 from CO_3^{2-}
(c) Shake two spatula endful of X with about 5cm^3 of water and filter. Keep both the filtrate and the residue. Divide the filtrate into six parts.	White residue stained blue. A pale blue filtrate	Non transition metal ion, Cu^{2+}
(i) To the first part of the filtrate add dilute sodium hydroxide solution drop-wise until in excess.	Pale blue precipitate insoluble in excess	Cu^{2+}
(ii) To the second part of the filtrate add dilute ammonia solution drop-wise until in excess.	Pale blue precipitate soluble in excess forming a deep blue solution	Cu^{2+}

(iii) To the third part of the filtrate carry out a test of your choice to confirm one of the cation in X. TEST: Potassium hexacyanoferrate (II) solution is added	A brown precipitate	Cu^{2+} confirmed
(iii) To the fourth part of the filtrate add zinc granules and warm (iv) Add Potassium iodide solution followed by sodium thiosulphate	Pale blue solution turn colourless and brown solid deposited White precipitate in brown solution Brown colour is discharged leaving a white precipitate	Cu^{2+} reduced to Cu Cu^{2+} forms Cu_2I_2 I_2 reduced to I^-
(v) To the fifth part of the filtrate add 2-3 drops of lead (II) nitrate solution. (vi) Two drops of concentrated H_2SO_4 and heat	No observable change Sweet fruity smell	SO_4^{2-} or Cl^- absent Ester formed, CH_3COO^- present

(vii) Use the sixth part of the filtrate to carry out a test of your own choice to confirm one of the anions in X. FeCl₃ solution is added and the mixture heated	Brown precipitate	CH ₃ COO ⁻ confirmed
(c) Wash the residue with water, add dilute hydrochloric acid drop wise to just dissolve the residue. Divide the solution into five parts.	Colourless solution is formed	Zn ²⁺ Al ³⁺ , Ca ²⁺ , Mg ²⁺ , Ba ²⁺
(i) To the first part of the solution add dilute sodium hydroxide solution drop wise until in excess.	White precipitate soluble in excess forming a colourless solution	Zn ²⁺ , Al ³⁺
(ii) To the second part of the solution add dilute ammonia solution drop wise until in excess.	White precipitate soluble in excess forming a colourless solution.	Zn ²⁺
(iv) Use the fourth part of the filtrate to carry out a test of your own choice to confirm another cation in X. Solid Ammonium chloride is added followed by disodium hydrogen phosphate and then excess aqueous ammonia	White precipitate soluble in excess ammonia	Zn ²⁺ confirmed

The cations in **X** are Cu^{2+} and Zn^{2+}

The anions in **X** are CO_3^{2-} and CH_3COO^-

3. You are provided with an organic compound **T** and you are required to identify the nature of compound **T**. Carry out the following tests on the compound and record your observations and deductions in the table below.

TEST	OBSERVATIONS	DEDUCTIONS
(a) Burn a little of T on a spatula end.	T burns with a yellow non sooty flame	Saturated aliphatic organic compound of low carbon content
(b) Add sodium hydroxide solution to a little of T in a test tube and shake well.	White solid dissolves forming a colourless solution	Carboxylic acid
(c) To a little of T in a test tube add about 5cm^3 of water and shake well. Test the mixture with litmus paper. Divide the mixture into five parts.	White solid dissolves forming a colourless solution. Blue litmus paper turns red	Polar organic compound of low molar mass Carboxylic acid
(i) To the first part, add sodium carbonate solution.	Effervescence occurs (bubbles)	Carboxylic acid confirmed
(ii) To the second part, add neutral iron (III) chloride solutions.	No observable change	Phenol absent
(iii) To the third part add Brady's reagent.	No observable change	Carbonyl compound absent
(iv) Add acidified potassium dichromate and heat	Orange soln turns to green	Primary alcohol or secondary alcohol
(v) To the fourth part add three drops of concentrated sulphuric acid followed by ethanoic acid and warm. Pour the mixture in cold water.	Sweet fruity smell	ester formed T is primary alcohol or secondary alcohol an alcohol
(vi) To the fifth part add an equal volume of ethanol and then three drops of concentrated sulphuric acid and warm. Pour the mixture in cold water.	Sweet fruity smell	Easter formed T is a carboxylic acid

(vii) Add soda lime and heat.	Bubbles of a colourless gas turns blue litmus paper and lime water milky	Decarboxylation of carboxylic acid
(viii) To the sixth part add acidified potassium manganate solution and heat. Divide the solution into two: - add Brady's reagent - add Tollens reagent	Purple solution turns colourless Yellow precipitate is formed No silver mirror	Primary alcohol oxidized to aldehyde or secondary alcohol oxidized to ketone Aldehyde absent So secondary alcohol oxidized to ketone Aldehyde absent
(ii) To the seventh part add Lucas reagent.	Cloudy solution is formed in 8min	Secondary alcohol

Comment on the nature of T

T is a saturated aliphatic secondary alcohol with a carboxyl group

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CONFIDENTIAL

Each candidate should be provided with the following;

1 pipette 10ml

1 measuring cylinder 50 or 100ml

1 measuring cylinder 10ml

2 conical flasks

1 empty beaker

Solid P which potassium dichromate (VI)

100ml of FA1

120ml of FA2

1.2g of P

3g of X

3g of T

8 test tubes in a rack

FA1 is 0.1M sodium thiosulphate

FA2 is 3.16g/l KMnO_4

X is $\text{ZnCO}_3 + (\text{CH}_3\text{COO})_2\text{Cu}$ in a ratio of 2:1

T is citric acid

